Arnav Patil

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Education

University of Toronto Sept 2023 – Apr 2028
BASc in Computer Engineering with PEY Co-op *Toronto, ON*

- **GPA: 3.63/4.00** with recognition on Dean's Honours List
- **Minor in Engineering Business**

Experience

Digital Design Engineer Intern May 2026 – Present
Marvell Technology *Toronto, ON*

- Building SystemVerilog models of analog IP blocks, cutting simulation runtime while matching SPICE results
- Generating simulation flows integrating SPICE and Verilog models to verify digital-analog interactions for D2D IP
- Automating Verilog generation with YAML- and Excel-based scripts, saving 50+ designer-hours per project

FPGA Engineer Intern May 2025 – Aug 2025
Intelligent Sensory Microsystems Lab *Toronto, ON*

- Developed FPGA-based modules for novel CMOS time-of-flight sensors in Prof. Roman Genov's **ISML** group
- Achieved 800 Mb/s sensor readout throughput by designing high-speed RTL pipelines on Xilinx Artix-7 FPGAs
- Resolved clock-domain crossing hazards using synchronizer circuits to rectify screen tearing and visual interference

Teaching Assistant Sept 2025 – Dec 2025
University of Toronto *Toronto, ON*

- Led tutorials for **APS100 Orientation to Engineering**, reinforcing engineering concepts through discussions
- Mentored first-year engineering students on academic planning, study strategies, and professional development

Projects

MOSAIC: Multi-Objective Smooth AnalytICal Placer |

- Macro placement for the **2026 Macro Placement Challenge** co-hosted by Partcl and Hudson River Trading
- Achieved a 10.3% improvement over RePlAce on proxy cost across IBM ICCAD04 benchmarks at 99 s/benchmark
- Implemented differentiable approximations of TILOS MacroPlacement functions for gradient-based optimization

SpecNet: Deep Learning for Analog Circuit Design |

- Developed an ML pipeline that predicts transistor sizing parameters from target analog circuit specifications
- Achieved 98% accuracy within 5% of optimal dimensions, greatly reducing number of SPICE simulations required
- Automated data generation by parsing LTspice simulation outputs into structured datasets for supervised learning

DE1-SoC Arcade Game |

- Implemented a blackjack game in Verilog on an FPGA, managing FSMs for game logic and player interaction
- Simplified hardware peripheral control from top-level modules by writing API-like wrapper RTL over Altera IP
- Fixed version compatibility issues in provided IP by writing a Python script to re-format memory files (`.mif`)

Nios-V Ultrasound System |

- Integrated an ultrasonic sensor and servo motor into the FPGA-based Nios-V soft processor via GPIO ports
- Designed & implemented BJT pull-up/pull-down networks for safe voltage conversion between FPGA and sensor
- Implemented timer-based polling in C to calculate object distance from 20 cm to 2 m within 2 cm precision

Skills & Coursework

Languages: Verilog/SystemVerilog, C/C++, Bash, Python, RISC-V assembly, Tcl
Tools: Synopsys VCS, Cadence Virtuoso, AMD Vivado, Intel Quartus Prime
Hardware: Computer Architecture, Digital & Analog Electronics, Signal Processing
Software: Operating Systems, Embedded Systems, Machine Learning